

Low-degree classification of torus-equivariant Keller maps in dimension three *

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Abstract

Motivated by the recent counterexample to the Jacobian conjecture in dimension three announced by Alpöge and credited to the AI system Fable [1], which is homogeneous for the $\mathbb{C}^*$ -action with weights $(1, -1, -2)$ and has polynomial degree seven, we classify all Keller maps compatible with the same torus action in polynomial degree at most four: every $(1, -1, -2)$ -equivariant Keller map of degree at most four is a polynomial automorphism (Theorem 2). The classification was first established computationally in conditional form and became unconditional on 2026-07-21 once the decomposition was certified exhaustive in two independent computer algebra systems (Section 5). The question of whether the same holds in degrees five and six — equivalently, whether degree seven is minimal within this symmetry class — is posed as an open computational problem, and the degree-five system (13 unknowns, 17 equations) is supplied.

1 Introduction

A polynomial map $F: \mathbb{C}^n \rightarrow \mathbb{C}^n$ is a *Keller map* if $\det \operatorname{Jac} F$ is a nonzero constant. The Jacobian conjecture [3, 4] asserted that every Keller map is a polynomial automorphism; it is now false for $n \geq 3$ by the explicit counterexample of [1]. The verification paper [2] checks, in exact arithmetic, that the counterexample’s Jacobian determinant is a nonzero constant and that the map is not injective (an explicit multi-point fiber is exhibited), and derives a binary-cubic description of the fibers, of generic cardinality three; the present note relies only on these verified facts about the example. The counterexample has polynomial degree seven and is homogeneous with respect to the grading $\operatorname{wt}(x, y, z) = (1, -1, -2)$, with component weights $(-2, -1, 1)$.

A natural first question about the mechanism is whether it is available in lower polynomial degree within the same symmetry class. This note answers that question negatively for degree at most four (Theorem 2), first in conditional form and, as of 2026-07-21, unconditionally once the reduction to a single certification task (Section 5) was discharged.

We emphasize the distinction between three evidentiary levels used throughout:

L1 *Proved by hand*: the equivariant ansatz and the forced-normalization lemma (Lemma 1).

*Draft. Computational classification, unconditional as of 2026-07-21 (Level-3 certification, Section 5, discharged; see the status box after the title for artifacts). All certificates, solver transcripts, and reproduction files accompany this note.

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L2 *Exactly machine-certified:* Jacobian-constancy identities, explicit inverse formulas, composition identities, denominator conditions, and the $b_0 = 0$ stratum, with certificates as symbolic objects over exact fields (no floating point).

L3 *Discharged 2026-07-21:* certified exhaustive decomposition of the Keller ideal, cross-checked in two independent systems (Section 5). The main theorem is unconditional.

2 The equivariant family and forced normalization

Fix the $\mathbb{C}^*$ -action $t \cdot (x, y, z) = (tx, t^{-1}y, t^{-2}z)$ and require $F = (F_1, F_2, F_3)$ to have components of weights $(-2, -1, 1)$, the weights of the known counterexample. For a degree bound D , write each F_i as a linear combination of the monomials of the prescribed weight and total degree at most D .

Lemma 1 (Forced normalization). *The only degree-one monomials of weights $-2, -1, 1$ are z, y, x respectively, so the linear part of any such F is anti-diagonal, $(\alpha z, \beta y, \gamma x)$, and $\det \text{Jac } F(0) = -\alpha\beta\gamma$. If F is Keller then $\alpha\beta\gamma \neq 0$, and F may be normalized to $\alpha = \beta = \gamma = 1$ by equivariant diagonal linear changes of source and target. Hence the normalized family considered below contains every $(1, -1, -2)$ -equivariant Keller map of degree at most D up to such changes.*

Proof. Weight bookkeeping gives the monomial claim; the constant $\det \text{Jac } F$ equals its value at the origin, which is the determinant of the linear part. Diagonal linear maps $\text{diag}(s, t, u)$ commute with the torus action, so pre- and post-composition by such maps preserves the equivariance class; the six available scalings (with a one-dimensional redundancy from the torus itself) normalize the three anchor coefficients to 1. $\square$

For $D = 4$ the normalized family is explicitly

$$F = (z + a_0y^2 + a_1xyz + a_2xy^3 + a_3x^2z^2, y + b_0xz + b_1xy^2 + b_2x^2yz, x + c_0x^2y + c_1x^3z),$$

nine unknown coefficients; imposing constancy of $\det \text{Jac } F$ yields 11 polynomial equations over $\mathbb{Q}$ (the *Keller ideal*). Note that for weight-homogeneous F the determinant is automatically of weight zero, i.e. a polynomial in the invariants xy and x^2z ; this is the structural reason the constancy condition is a manageable system.

3 Certification methodology (L2)

The Keller ideal was decomposed with SymPy 1.14.0 (Python 3.12.3) via Gröbner-based triangular decomposition (`sympy.solve`). The raw case-split returns 27 branches, several re-derived along redundant elimination paths; deduplicating by solution locus leaves 5 distinct parametrized families. (Correction, 2026-07-21: earlier circulated drafts of this note quoted the raw, undeduplicated branch count; no family or certificate was lost by deduplicating, only redundant re-derivations of the same locus.) The completeness of this decomposition was item L3, discharged 2026-07-21 (Section 5). For each family, with free parameters kept *symbolic*:

1. the substituted determinant is expanded and verified to be a constant identically in (x, y, z) (it equals -1 on every family);

2. a lexicographic Gröbner basis of $\langle F_1 - p, F_2 - q, F_3 - r \rangle$ ($x > y > z$) is computed over the exact function field $\mathbb{Q}(\text{parameters})(p, q, r)$, the unique solution $(x, y, z) = (X, Y, Z)(p, q, r)$ is extracted, and the composition identities $F_i(X, Y, Z) = p, q, r$ are verified by exact cancellation;
3. all irreducible factors of all denominators of X, Y, Z are computed. If every factor is a nonzero constant, the inverse is valid for *every* parameter specialization in the family's domain (*pointwise* certificate); otherwise the factors are recorded as exceptional.

A rational inverse with polynomial re-composition to the identity makes the map birational; a birational Keller map is a polynomial automorphism. Indeed, a Keller map is étale, hence quasi-finite, and a separated, quasi-finite, birational morphism onto a normal variety is an open immersion by Zariski's main theorem [6]; applied to $F: \mathbb{C}^n \rightarrow \mathbb{C}^n$ this makes F injective, and an injective polynomial self-map of $\mathbb{C}^n$ is an automorphism with polynomial inverse [5]. When the certificate is pointwise, this conclusion holds at every parameter value, not only generically.

Outcome: 4 of 5 distinct families certify pointwise. The remaining 1 certifies with the single exceptional factor b_0 (the coefficient of xz in F_2), hence pointwise on the constructible piece $b_0 \neq 0$, which is the entire stated domain of that parametrization. The complementary constructible piece is handled by decomposing the ideal augmented by b_0 : after the same deduplication it yields 1 distinct family, which certifies pointwise. (Second correction, 2026-07-21: this stratum's decomposition initially appeared to yield 2 distinct families, for 7 total; one of those two turned out to be an exact duplicate of a *main*-branch family — the main decomposition, not told to assume $b_0 = 0$, had independently landed on a branch already sitting there. Deduplicating across both files, not only within each, leaves 6 distinct families total, 5 from the main piece and 1 genuinely new from the stratum.) All certificates are supplied as machine-readable JSON with the full inverse formulas. An independent verifier — a separate program that reads only the static certificates and checks the determinant identities, both compositions $F \circ H = \text{id}$ and $H \circ F = \text{id}$, and the recorded denominator factors, performing no solving — passes on all 6 distinct families. The ideal-theoretic certification (Section 5) separates the two loci as the saturation $I : b_0^\infty$ and the sum $I + \langle b_0 \rangle$ and requires their components to be covered by the certified families – carried out and discharged 2026-07-21.

4 Main result

Theorem 2 (Computational classification). *Every $(1, -1, -2)$ -equivariant Keller map $\mathbb{C}^3 \rightarrow \mathbb{C}^3$ of polynomial degree at most four is a polynomial automorphism.*

The Keller ideal I has exactly three minimal primes (Singular 4.4.1 `minAssGTZ`, independently reproduced in Macaulay2 1.26.06 `decompose`). Minimal primes suffice because the theorem is a statement about *points*: the set of normalized equivariant Keller maps is the affine variety $V(I)$, and set-theoretically $V(I) = V(P_1) \cup V(P_2) \cup V(P_3)$ over the minimal primes, irrespective of any embedded associated primes, which contribute no additional points (no radicality claim about I is made or needed). The relation between the six certified families and the three components is as follows (`out/certify.log`): families 0, 2, and 5 are dense charts of the three components — component 1 $\leftrightarrow$ family 5, 2 $\leftrightarrow$ 2, 3 $\leftrightarrow$ 0 — with family 5 the b_0 -stratum family, so one entire component lies inside $\{b_0 = 0\}$; the remaining families 1, 3, 4 are constructible subfamilies contained in these closures, which arose as separate branches of the triangular decomposition and are retained with their own certificates. Every point of every component carries an explicit

polynomial inverse with both compositions verified: the pointwise certificates are valid on their entire loci, and the single non-pointwise chart (family 0, valid on $b_0 \neq 0$) is complemented by the $b_0 = 0$ stratum decomposition (Section 3).

Remark 3. *The theorem should be read together with Lemma 1: the normalization is not a hypothesis but a reduction, so the conclusion covers the full equivariant class up to equivariant diagonal linear changes.*

Remark 4 (Certification history). *Earlier drafts stated Theorem 2 conditionally on the exhaustiveness of the computed decomposition; that certification is now discharged (Section 5). The certification process surfaced and corrected two family-count bookkeeping errors and two computer-algebra usage bugs, none affecting the mathematical content. The full audit trail — the family-count correction history and the specific Singular and Macaulay2 pitfalls, recorded so that others do not lose time to them — is kept in the accompanying repository (`HANDOFF.md`) rather than here.*

5 Certification history (L3, discharged 2026-07-21)

The single previously-uncertified step was exhaustiveness of the two decompositions. The accompanying files `certify_exhaustiveness.sing` (Singular, `minAssGTZ` plus containment checks against the 6 certified family ideals) and `deg4.decomposition.m2` (Macaulay2, `decompose`) compute the minimal associated primes of the Keller ideal over $\mathbb{Q}$. Both agree on 3 minimal primes; every one is covered by a certified family, and every certified family lies inside the Keller variety (both containment directions checked, see `out/certify.log`). The theorem is therefore unconditional: no omitted component remains that could contain maps outside the certified families.

6 Open problems

Problem 5. *Decide degrees five and six. The degree-five Keller system (13 unknowns, 17 equations, supplied in `deg5.system.txt`) exceeded the pure-Python Gröbner engine used here; an exact decomposition in Singular is running as of this writing (2026-07-21) but has not finished, apparently limited by intermediate coefficient growth rather than the size of the search space — three independent modular runs (primes 32003, 32009, 65003) each find exactly 2 minimal primes with identical structure, and three exact families reconstructed from them (one splitting into two further branches) each independently verify: $\det = -1$ and an explicit polynomial inverse with verified composition, by the same method as the certified degree-4 families (`04-scripts/reconstruct_deg5_candidates.py`). **This is not a classification result for degree five** — the modular component count is a heuristic preview, not a characteristic-zero certificate, and exhaustiveness of these families has not been established by any method. A negative answer through degree six would establish degree seven as minimal within this symmetry class and invite a conceptual explanation, possibly via the binary-cubic fiber mechanism of [2].*

Problem 6. *Enumerate all weight systems (w_1, w_2, w_3) admitting nontrivial equivariant Keller families in low degree, and run the same classification across them.*

Problem 7. *Replace the branch-by-branch computation with a structural proof, e.g. a triangular factorization theorem for low-degree equivariant Keller maps.*

Acknowledgment

The formulation and the exploratory symbolic computations were carried out with the assistance of Claude Fable (Anthropic). All results are certified by machine-checkable artifacts in exact arithmetic — certificates, solver transcripts, and an independent solving-free verifier in the accompanying repository — independent of any AI system’s reasoning and reproducible from the supplied materials. Correspondence: minustsquared@gmail.com.

References

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